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Assignment topic:	Security Mechanism
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Q. Explain Security Mechanism and its types.Briefly?

Security Mechanism

Definition

A device or function designed to provide one or more security services usually rated in terms of strength of service and assurance of the design.

OR

A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.

Types of Security Mechanism

- 1. Cryptography:** It is the study about protocols that prevent third party from reading private messages.It is also about constructing and analyzing protocols for secure communication.
- 2. PKI (Public key Infrastructure):** It is set of rules or policy that are used to create, manage, distribute, store, use and revoke digital certificates and manage public key encryption.
- 3. Digital Signature:** This security mechanism is achieved by adding digital data that is not visible to eyes. It is form of electronic signature which is added by sender which is checked by receiver electronically. This mechanism is used

to preserve data which is not more confidential but sender's identity is to be notified.

- 4. Encipherment:** This security mechanism deals with hiding and covering of data which helps data to become confidential. It is achieved by applying mathematical calculations or algorithms which reconstruct information into not readable form. It is achieved by two famous techniques named Cryptography and Encipherment. Level of data encryption is dependent on the algorithm used for encipherment.
- 5. Access Control :** This mechanism is used to stop unattended access to data which you are sending. It can be achieved by various techniques such as applying passwords, using firewall, or just by adding PIN to data.
- 6. Notarization :** This security mechanism involves use of trusted third party in communication. It acts as mediator between sender and receiver so that if any chance of conflict is reduced. This mediator keeps record of requests made by sender to receiver for later denied.
- 7. Data Integrity :** This security mechanism is used by appending value to data to which is created by data itself. It is similar to sending packet of information known to both sending and receiving parties and checked before and after data is received. When this packet or data which is appended

is checked and is the same while sending and receiving data integrity is maintained.

8. Authentication exchange : This security mechanism deals with identity to be known in communication. This is achieved at the TCP/IP layer where two-way handshaking mechanism is used to ensure data is sent or not

9. Bit stuffing : This security mechanism is used to add some extra bits into data which is being transmitted. It helps data to be checked at the receiving end and is achieved by even parity or odd parity.
